On Secret Multiple Image Sharing

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Abstract

In traditional secret sharing schemes for general access structures, a secret can be encoded into n sets of (shadow, key)'s which are distributed to n participants one by one. Only those participants belonging to a pre-defined qualified subset can recover the shared secret by using their (shadow, key)'s. The size of (shadow, key)'s may be different for different participants. This problem exists when the traditional secret sharing schemes are adopted to cope with the sharing of multiple secret images for general access structures. Specifically, one participants may hold more than one set of (shadow, key)'s. In this paper, we design a new multiple secret images sharing scheme for both of the r out of n threshold and general access structures. Our scheme produces n keys with a constant size, which are distributed to the n participants, and some shadows which are stored in public. Each participant in our scheme holds only a constant-size key. The burden for each participant to carry one or more shadow(s) in traditional secret multiple images sharing schemes can be removed

Keywords: secret sharing, shadow, threshold scheme, general access structure, qualified subset.

1. Introduction

A *secret sharing scheme* is able to protect and share a secret among a group of n participants. It encodes a secret into n parts of information called *shadows* and *keys*. Each participant holds his own set of (shadow, key) privately which leaks nothing about the secret. To recover the secret, an *access structure* can be specified which defines the groups of participants that are feasible to recover the shared secret by their (shadow, key)'s. When the feasible access condition is designated to be "any group of r participants", we call it an r out of n, denoted by (r, n) *threshold* access structure.

Shamir [8] in 1979 proposed a simple scheme to accomplish the (r, n) threshold secret sharing. Consider an $r-1$ degree polynomial:

$$f(x) = a_0 + a_1x^1 + a_2x^2 + \ldots + a_{r-1}x^{r-1}$$

where all computations are perform in $GF(p)$ in which p is a prime, $1 \le a_{r-1} < p$, $0 \le a_z < p$ for $0 \le z \le r-2$, and $1 \le x < p$. Shamir's (r, n) scheme utilizes this polynomial to share a secret s as follows. The dealer sets s to be a_0 and randomly chooses $a_1, a_2, \ldots, a_{r-1}$ to form $f(x)$. Then, he determines $x_1, x_2, \ldots, x_n$ (as *keys*) based upon which $f(x_1), f(x_2), \ldots, f(x_n)$ are computed (as *shadows*). The n pairs of $(f(x_i), x_i)$'s, $1 \le i \le n$, are distributed to the n participants one by one. Since any group of r (or more) $(f(x_i), x_i)$'s is able to compute $(a_0, a_1, \ldots, a_{r-1})$ by solving the r equations simultaneously, s $(=a_0)$ is thus recovered. None of any group of less than r participants can solve the r equations completely. We say that s is shared by n participants in an (r, n) threshold structure.

Thien and Lin [11] in 2002 extended Shamir's scheme in the area of secret image sharing. Consider an image P with N pixels in total to be shared in an (r, n) threshold structure. They diffuse all N pixels of P and organize them into N/r segments with r pixels each. For each segment of r pixels, say segment t $(1 \le t \le N/r)$, which is assigned to be $(a_0, a_1, \ldots, a_{r-1})_t$, i.e. the r coefficients of the above formula, to form $f_t(x)$. The dealer determines n keys $x_1, x_2, \ldots, x_n$, and computes $f_t(x_1), f_t(x_2), \ldots, f_t(x_n)$ for $1 \le t \le N/r$. Then, $f_1(x_i), f_2(x_i), \ldots, f_{N/r}(x_i)$ are merged into a *shadow image* D_i for $1 \le i \le n$. The dealer gives (D_i, x_i) to participant i for $1 \le i \le n$. It is not hard to see that only r (or more) participants can recover $(a_0, a_1, \ldots, a_{r-1})_t$ for all equations $f_t(x)$'s, $1 \le t \le N/r$ by their r (D_i, x_i)'s, which reconstruct all pixels of P accordingly. The shadow size of Thien and Lin's approach is N/r, that is, D_i contains N/r pixels for $1 \le i \le n$. If the original Shamir's scheme is directly applied to share an image, the size of each shadow is N.

Secret sharing schemes for general access structures were discussed in [1, 5, 10]. Most of these researches were based upon *Shamir*'s scheme. All of the schemes mentioned above (either for threshold or for general access structures) aim at the sharing of one secret (which is a number or an image).

The problems about sharing multiple secrets were discussed in [4, 6, 7, 11, 12]. The skills adopted in these researches include Shamir's polynomial-based approach [7, 11], XOR operation [12] (only feasible for decoding multiple secrets from various pairs of participants), one-way function [6], relationship of sharing circle [4], multi-group and Chinese remainder theorem [7]. Some schemes do not realize general access structures, but instead, they deal with some restricted cases, such as sharing circle [4] or qualified subsets with size 2 [12]. Nevertheless, all these researches suffer from the fact that the sizes of (shadow, key)'s are different for various participants. Generally speaking, the more secrets a participant may share with others, the larger his/her shadow may be. If the size of the largest shadow and the participant holding it were wired, a direct threat is that the malicious intruders might simply ruin the largest shadow to devastate the recovery of some shared secrets corresponding to the shadow. This destroys the integrity of the seared multiple secrets.

In this paper, we devise a new secret sharing scheme for multiple secret images with the feature that each participant is distributed by a constant-size key only. The rest of the paper is organized as follows. We present the proposed schemes for threshold and general access structures respectively in Section 2. The experimental results of our schemes are summarized and discussed in Section 3. Section 4 gives some concluding remarks.

2. The proposed scheme

In the following we assume that q images are shared by n participants. Let $P = \{1, 2, \ldots, n\}$ be the set of these n participants and $Q = \{S_1, S_2, \ldots, S_q\}$ be the set of the q images where the size of S_j is $w_j \times h_j$ for $1 \leq j \leq q$. Note that the sizes of these q secret images may be different. The proposed schemes also exploit Shamir's polynomial-based approach; yet, all of the computations in our schemes are in $GF(2^8)$ which is addressed in Section 2.1, Sections 2.2 and 2.3 respectively present the threshold sharing scheme for multiple secret images, denoted by (r, P, Q)-SSMSI, and the general access structure sharing scheme for multiple secret images, denoted by (G, P, Q)-SSMSI.

2.1 GF(2^8)

We point out first that all computations performed in the proposed schemes for sharing multiple images are in $GF(2^8)$. In Thien and Lin's scheme, their computations are in $GF(251)$ since 251 is the largest prime no greater than 256 (the number of gray scales in a gray level image). Many following works adopted the same setting of $p = 251$ [4, 12]. In fact, choosing $p = 256$ ($=2^8$) and taking all computations in $GF(2^8)$ are perfectly for dealing with gray level images. Please refer to [9, 13] for detailed descriptions about the computations under $GF(2^8)$.

2.2 (*r, P, Q*)-SSMSI

(r, P, Q)-SSMSI is defined as a scheme that is able to share q secret images in Q among n participants in P such that r ($r \leq n$) participants can reconstruct all these q secret images. In the encoding and decoding processes we shall decompose each image into segments of d pixels. The encoding and the decoding schemes of our (r, P, Q)-SSMSI are illustrated as follows.

• Encoding scheme of (r, P, Q)-SSMSI:

Input: $P = \{1, 2, \ldots, n\}$, $Q = \{S_1, S_2, \ldots, S_q\}$, r (threshold), and d (segment size)

Output: q encrypted shadows $H_1, H_2, \ldots, H_q$, and n pairs of keys (x_i, y_i)'s for $1 \leq i < n$

1. The dealer randomly chooses r integers a_0, $a_1, \ldots, a_{r-1}$, ($1 \leq a_{r-1} < p$, $0 \leq a_z < p-1$ for $0 \leq z \leq r-2$). These r integers are assigned as the coefficients of the (r–1)-degree polynomial $f(x)$ as follow:

   $$f(x) = a_0 + a_1x^1 + a_2x^2 + \ldots + a_{r-1}x^{r-1}$$

2. The dealer chooses n different keys $x_1, x_2, \ldots, x_n$ ($1 \leq x_i < p$ for $1 \leq i \leq n$) to construct y_i by computing $y_i = f(x_i)$ for $1 \leq i \leq n$.
3. The dealer uses a seed to generate a permutation sequence to permute the pixels of the secret images S_j to S_j', $1 \leq j \leq q$.
4. The dealer computes $g_j^a(x)$ with respect to S_j' by segments of d pixels in S_j' for $1 \leq a \leq l_j$ ($=w_j \times h_j/d$) as follow:

   $$g_j^1(x) = s_j^{1,1} + s_j^{1,2}x^1 + \ldots + s_j^{1,d}x^{d-1}$$
   $$g_j^2(x) = s_j^{2,1} + s_j^{2,2}x^1 + \ldots + s_j^{2,d}x^{d-1}$$
   $$\ldots$$
   $$g_j^{l_j}(x) = s_j^{l_j,1} + s_j^{l_j,2}x^1 + \ldots + s_j^{l_j,d}x^{d-1}$$

   There are l_j polynomials and $s_j^{a,b}$ is the bth pixel in segment a.
5. The dealer computes $f(x)$ and $g_j^a(x)$ to get polynomial $h_j^a(x) = f(x) \times g_j^a(x)$ where $1 \leq j \leq q$, $1 \leq a \leq l_j$.
6. All coefficients in $h_j^a(x)$'s for $1 \leq \alpha \leq l_j$ are merged into an encrypted shadow H_j for $1 \leq j \leq q$.

7. The dealer puts q encrypted shadows H_1, H_2, …, H_q on a public space. The key (x_i, y_i) is distributed to participant i for $1 \le i \le n$.

Note that the above computations are for gray level images. Regarding to color images, we apply the above computations with respect to the R, G and B components respectively of each pixel in the color image.

• Decoding scheme of (r, P, Q)-SSMSI:

Input：r participants $i_1, i_2, \ldots, i_r$, and d (segment size)

Output：q secret images $S_1, S_2, \ldots, S_q$

1. All participants i_k's for $1 \le k \le r$ use their keys (x_{i_k}, y_{i_k})'s to find out the coefficient a_z's for $0 \le z \le r-1$ in $f(x)$ as follows:

$$f(x) = \sum_{j=i_1}^{i_r}\left(y_j \prod_{u=i_1, u\neq j}^{i_r} \frac{x - x_u}{x_j - x_u} \right)$$

$$(= a_0 + a_1x^1 + a_2x^2 + \ldots + a_{r-1}x^{r-1})$$

2. After accessing H_j from the public space, the participants form $h_j^a(x)$'s by decomposing H_j into segments of $r+d-1$ pixels for $1 \le a \le l_j$ and $1 \le j \le q$.
3. The participants obtain $g_j^a(x)$ via $h_j^a(x)$ and $f(x)$ to as follow:

$$g_j^a(x) = h_j^a(x)/f(x)$$
$$= s_j^{a,1} + s_j^{a,2}x^1 + \ldots + s_j^{a,d}x^{d-1}$$

for $1 \le a \le l_j$ and $1 \le j \le q$.

4. All coefficients in $g_j^a(x)$ can be merged into an image S_j' for $1 \le a \le l_j$ and $1 \le j \le q$.
5. The participants apply the inverse- permutation operation to the permuted image S_j' to get the secret image S_j.
6. These q images S_1, S_2, …, S_q can be reconstructed by the r participants.

2.3 (G, P, Q)-SSMSI

Let $G = (\boldsymbol{A}, \boldsymbol{B})$ be an access structure for sharing multiple secrets where $\boldsymbol{A} = \{A_1, A_2, \ldots, A_m\}$ is the set of m qualified subsets of P and $\boldsymbol{B} = \{B_1, B_2, \ldots, B_m\}$ is the set of m subsets of Q such that A_t, say $A_t = \{i_1, i_2, \ldots, i_{ut}\}$ can reconstruct secret images in B_t, say $B_t = \{S_{j_1}, S_{j_2}, \ldots, S_{jn_t}\}$ for $1 \le t \le m$. Given G, P and Q, the secret sharing scheme for multiple secrtes in Q under G is denoted as (G, P, Q)-SSMSI.

• Encoding scheme of (G, P, Q)-SSMSI:

Input: $P = \{1, 2, \ldots, n\}$, $Q = \{S_1, S_2, \ldots, S_q\}$, $G = (\boldsymbol{A}, \boldsymbol{B})$, and d (segment size) where $A_t = \{i_1, i_2, \ldots, i_{ut}\} \in \boldsymbol{A}$, $B_t = \{S_{j_1}, S_{j_2}, \ldots, S_{jn_t}\} \in \boldsymbol{B}$ $(1 \le t \le m)$

Output: $m \times q$ encrypted shadows $H_{t,j}$, and n pairs of keys (x_i, y_i)'s

1. The dealer chooses n pairs of random and non-repeat integers (x_i, y_i)'s as keys for $1 \le x_i < p$, $0 \le y_i < p$.
2. For each group A_t with m_t participants in $\boldsymbol{A}$ ($1 \le t \le m$), the dealer uses the keys of these m_t participants (in A_t) to construct $f_t(x)$ of degree m_t-1 by Lagrange's interpolation as follow:

$$f_t(x) = \sum_{j=i_1}^{i_{m_t}}\left(y_i \prod_{k=i_1, k\neq j}^{i_{m_t}} \frac{x - x_k}{x_j - x_k} \right)$$

$$(=a_0 + a_1x^1 + a_2x^2 + \ldots + a_{m_t-1}x^{m_t-1}).$$

3. The dealer chooses a seed to generate a permutation sequence to permute the pixels of the secret images S_j to S_j', $1 \le j \le q$.
4. For each S_j' in B_t, where $1 \le t \le m$, the dealer constructs g_j^a by segments of d pixels in S_j' for $1 \le a \le l_j$ $(= w_j \times h_j / d)$ as follows:

$$g_j^1(x) = s_j^{1,1} + s_j^{1,2}x^1 + \ldots + s_j^{1,d}x^{d-1}$$
$$g_j^2(x) = s_j^{2,1} + s_j^{2,2}x^1 + \ldots + s_j^{2,d}x^{d-1}$$
$$\ldots$$
$$g_j^{l_j}(x) = s_j^{l_j,1} + s_j^{l_j,2}x^1 + \ldots + s_j^{l_j,d}x^{d-1}.$$

There are l_j polynomials and $s_j^{a,b}$ is the *bth* pixel in segment a.

5. Compute $h_{t,j}^a(x) = f_t(x) \times g_j^a(x)$ where $1 \le t \le m$, $1 \le j \le q$, $1 \le \alpha \le l_j$.
6. All coefficients in $h_{t,j}^a(x)$'s for $1 \le \alpha \le l_j$ are merged into an encrypted shadow $H_{t,j}$ for $1 \le t \le m$, $1 \le j \le q$.
7. The dealer puts encrypted shadows $H_{t,j}$'s on a public space. The key (x_i, y_i) is distributed to participant i for $1 \le i \le n$.

• Decoding scheme of (G, P, Q)-SSMSI:

Input: A_t, $\{j_1, j_2, \ldots, j_{vt}\}$ and d (segment size) where $A_t = \{i_1, i_2, \ldots, i_{ut}\}$ such that the participants in A_t can recover the secrets in $S_{j_1}, S_{j_2}, \ldots, S_{jn_t}$

Output: $B_t = \{S_{j_1}, S_{j_2}, \ldots, S_{jn_t}\}$

1. All participants i_k's in A_t compute the coefficients a_z's for $0 \le z \le u_t-1$ by their keys (x_{i_k}, y_{i_k}). Then they find $f_t(x)$ with respect to A_t as follows:

$$f_t(x) = \sum_{j=i_1}^{i_{m_t}} \left(y_i \prod_{k=i_1, k \neq j}^{i_{m_t}} \frac{x - x_k}{x_j - x_k} \right)$$

$$(=a_0+a_1x^1+a_2x^2+\ldots+a_{m_t-1}x^{m_t-1})$$

2. Access $E_t = \{H_{t,1}, H_{t,2}, \ldots, H_{t,n_t}\}$ from the public space. For each $H_{t,j} \in E_t$ with N_t pixels in total for $1 \le j \le n_t$, $h_{t,j}^{\alpha}(x)$ can be derived for $1 \le \alpha \le l_j$ $(= N_t/(r+d-1))$. Each $h_{t,j}^{\alpha}(x)$ has $r+d-1$ pixels. Consequently,

$$g_j^{\alpha}(x) = h_{t,j}^{\alpha}(x)/f_t(x)$$

   can be found , and then the d pixels of segment α with respect to S_j' (i.e. $s_j^{\alpha,1}, s_j^{\alpha,2}, \ldots, s_j^{\alpha,d}$) are accordingly obtained.

3. All coefficients in $g_j^{\alpha}(x)$ for $1 \le \alpha \le l_j$ are merged into an image S_j'for $S_j' \in B_t$.

4. The corresponding participants apply the inverse-permutation operation to the permuted image S_j' to get the secret image S_j.

5. These n_t images $S_{j_1}, S_{j_2}, \ldots, S_{jn_t}$ can be reconstructed by participants in A_t.

3. Experimental results

Two experiments were designed to demonstrate the effectiveness of our schemes. Experiment 1 focuses on (r, P, Q)-SSMSI and Experiment 2 addresses on (G, P, Q)-SSMSI.

• Experiment 1. (r, P, Q)-SSMSI

Consider the case of $(r, P, Q) = (2, \{1, 2, 3, 4\}, \{S_1, S_2\})$ where S_1 and S_2 are chosen as the 512×512 colored Peppers and Leena images as shown in Figure 1. (a) and (b) respectively. The segment size is $d = 512$.

The dealer randomly chooses two $(r=2)$ integers, say 58 and 129, to be a_0 and a_1 in $f(x)$ which $f(x) = 58 + 129x$. Note that $0 \le a_0, a_1 \le 256$ and $a_1 \neq 0$. Then four $(n=4)$ different integers are selected as keys, say $x_1=1$, $x_2=2$, $x_3=3$ and $x_4=4$, based upon which y_i's $(=f(x_i)$'s) are computed so that four pairs of (x_i, y_i)'s: (1, 187), (2, 21), (3, 148), (4, 100) are obtained, which are distributed to the four participants in P one by one.

The dealer picks up a seed (=100, in this experiment) to generate a permutation sequence to permute the pixels of the secret images S_1 and S_1 into S_1' and S_2' respectively.

(a) (b) (c)

(d) (e) (f)

Figure 1. Implementation results of Experiment 1 (for (r, P, Q)-SSMSI): (a) S_1 (512×512), (b) S_2 (512×512); (c) H_1 (512×513) (d) H_2 (512×513); (e) reconstructed S_1 by any two participants in P, (f) reconstructed S_1 by any two participants in P.

There are l (= 512 = (512×512) / d where d = 512) segments in total. Let us show that how the first segment of d pixels is adopted in the encoding process for the first secret image S_1'.

We adopt these d pixels to compute $g_1^{1}(x)$, then $h_1^{1}(x) = f(x)\times g_1^{1}(x)$ can be obtained with degree 512 (= 1 + 511 = $(r - 1) + (d - 1)$) (containing 513 coefficients). Note that the numbers of coefficients in $f(x)$, $g_1^{1}(x)$, and $h_1^{1}(x)$ are 2, 512, and 513 respectively. All segments are adopted in the same way to produce $h_1^{a}(x)$'s accordingly for $1 \le a \le l$. The number of all coefficients in $h_1^{a}(x)$'s are 512×513 (= 512×l), $1 \le a \le l$, which is in the point of view of a gray level pixel. For a color images, the corresponding $h_1^{a}(x)$'s of R, G and B components are merged into H_1 with a size of 512×513 (color pixels). H_2 is produced likewise with respect to S_2. Figure 1. (c) and (d) show H_1 and H_2 with respect to $Q = \{S_1, S_2\}$.

The encrypted shadows H_1 and H_2 are placed in a public space. For any group of two participants, they can recover S_1 and S_2 by using their keys. They re-build $f(x) = 58 + 129x$ first via their keys, and then retrieve encrypted shadows H_1, H_2 from the public place. We explain how they perform the decoding process for the first segment of S_1.They get the first 513 values form H_1 to be the coefficients of $h_1^{1}(x)$, and compute $g_1^{1}(x) = h_1^{1}(x)/f(x)$. After all $g_1^{a}(x)$'s are thus found for $1 \le a \le l$, S_1' can be reconstructed. Then, the participants apply the inverse-permutation operation to the permuted image S_1' to get the secret image S_1.

Likewise, S_2 can be obtained in the same way. Figure 2. (e) and (f) show the recovered results which are exactly the same as S_1 (Figure 2. (a)) and S_2 (Figure 2. (f)).

• Experiment 2. (G, P, Q)-SSMSI

Consider the case of (G, P, Q) = ((({1, 2, 3}, {1, 4}),({S_1}, {S_1, S_2})), {1, 2, 3, 4}, {S_1, S_2}) where S_1 and S_2 are the same as in Experiment 1; and A_1 = {1, 2, 3} shares B_1 = {S_1} and A_2 = {1, 4} shares B_2 = {S_1, S_2}.

The dealer randomly chooses four (n = 4) pairs of keys (1, 131), (2, 210), (3, 206), (4, 112). Participants in group A_1 = {1, 2, 3} use their keys $(x_1, y_1) = (1, 131)$, $(x_2, y_2) = (2, 210)$, $(x_3, y_3) = (3, 206)$ to compute $f_1(x) = 159 + 120x + 100x^2$.

The dealer chooses a seed a seed (=100, in this experiment) to generate a permutation sequence to permute the pixels of the secret images S_1 and S_1 into S_1' and S_2' respectively. For the first secret segment of image S_1' in group B_1, $g_1^{1}(x)$, $h_{1,1}^{1}(x) = f_1(x)\times g_1^{1}(x)$ are accordingly found. The numbers of coefficients in $f_1(x)$, $g_1^{1}(x)$, and $h_{1,1}^{1}(x)$ are 3, 512, and 514. After all segments are processed, all coefficients in $h_{1,1}^{a}(x)$ are 514×512 for $1 \le a \le 512$ (=$(h_1\times w_1)/d$). The dealer merges all coefficients into an encrypted shadow $H_{1,1}$ which is 512×514 as shown in Figure 2. (a).

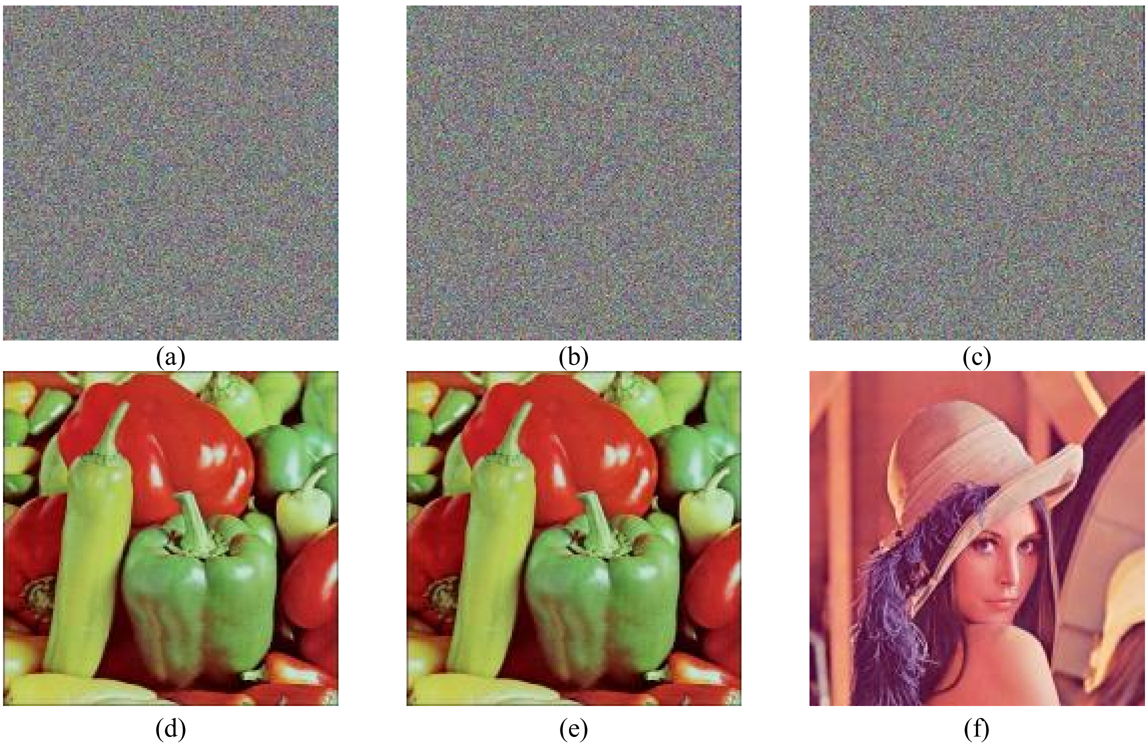

Figure 2. Implementation results of Experiment 2 (for (G, P, Q)-SSMSI). (a) $H_{1,1}$ (512×514), (b) $H_{2,1}$ (512×513), (c) $H_{2,2}$ (512×513); (d) reconstructed S_1 in B_1; (e) reconstructed S_1 in B_2; (f) reconstructed S_2 in B_2.

Participants in group $A_2 = \{1, 4\}$ use their keys $(x_1, y_1) = (1, 131)$, $(x_4, y_4) = (4, 112)$ to construct $f_2(x) = 87 + 212x$. Then, $g_1{}^1(x)$, $h_{2,1}{}^1(x) = f_2(x)\times g_1{}^1(x)$ and $H_{2,1}$ can be obtained where $H_{2,1}$ is 512×513 as shown in Figure 2. (b). By the same way, participants in the group A_2 can get $H_{2,2}$ which is 512×513 as shown in Figure 2.(c).

Participants in group A_t can construct $f_t(x)$ by their keys for $t = 1, 2$ in this experiment. They retrieve shadows $H_{t,j}$'s from the public space to recover the secrets. Figure 2. (d) shows the reconstructed secret image S_1 by A_1, while Figure 2. (e) and (f) show the reconstructed secret images S_1 and S_2 respectively by A_2.

4. Conclusion

The most significant contributions of this paper are that (1) the multiple secret images can be shared in threshold or general access structure; and (2) the each participant only holds a constant-size key. The shadows are placed in some public space. Without the legal combinations of keys, the shadows leak nothing about the secrets shared.

Since the size of the key is constant, the risk that a participant with large-size shadow may be attacked by malicious intruders in the traditional secret sharing schemes for multiple secrets can be reduced. This also implies that the participant is easies to carry a small key than a large shadow (in the traditional secret sharing schemes). He may even memorize the key to avoid the accident that the key may be lost.

Our schemes provide a new ideal for secret sharing scheme. Some extensions in our scheme can be further discussed, such as applying a dynamic segment size, using other operations instead of multiplications to produce shadows, to name a few.

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